

02_mlp_classifier - Results and Figures

Source: notebooks/02_mlp_classifier.ipynb

Results from cell 1

Text output:

```
Removendo linhas incompletas CARDT (dlc<8): 1189537
   id  val0  val1  val2  val3  val4  val5  val6  val7  flag
0  790    5    33   104    9    33    33    0   111  BENIGN
1  399   254    91    0    0    60    0    0    0  BENIGN
2  608    25    33    34   48    8   142   109   58  BENIGN
3  672   100    0   154   29   151    2   189    0  BENIGN
4  809    64   187   127   20    17    32    0   20  BENIGN
flag
BENIGN    15261030
DoS        662184
RPM        654897
GEAR       597252
Fuzzy     491847
Name: count, dtype: int64
```

Results from cell 3

Text output:

```
flag
BENIGN    15261030
DoS        662184
RPM        654897
GEAR       597252
Fuzzy     491847
Name: count, dtype: int64
```

Results from cell 5

Text output:

Train shape: (14133768, 9), Test shape: (3533442, 9)

Results from cell 9

Text output:

```
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
  warnings.warn(
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
  warnings.warn(
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
  warnings.warn(
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
  warnings.warn(
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
  warnings.warn(
```

[illegible]

```

yet.
warnings.warn(
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
warnings.warn(
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
warnings.warn(
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
warnings.warn(
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
warnings.warn(
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
warnings.warn(
c:\Users\thelo\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn\normal_network\_multilayer_per
ceptron.py:785:
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1) reached and the optimization hasn't converged
yet.
warnings.warn(

```

Text output:

	precision	recall	f1-score	support
BENIGN	0.99930	0.93823	0.96780	200000
DoS	0.91551	0.95499	0.93483	138674
Fuzzy	0.99883	0.97594	0.98725	115937
GEAR	0.88145	1.00000	0.93699	127483
RPM	0.99999	0.94079	0.96949	139275
accuracy			0.95892	721369
macro avg	0.95902	0.96199	0.95927	721369
weighted avg	0.96242	0.95892	0.95947	721369

Results from cell 10

Text output:

Treinando XGBoost em 400000 amostras de treino e avaliando em 80000 amostras de teste.

```

itera??o 10/50 test_loss=0.5492 test_acc=0.9541
itera??o 20/50 test_loss=0.2724 test_acc=0.9551
itera??o 30/50 test_loss=0.1715 test_acc=0.9585
itera??o 40/50 test_loss=0.1331 test_acc=0.9590
itera??o 50/50 test_loss=0.1166 test_acc=0.9593

```

XGBoost classification report

	precision	recall	f1-score	support
BENIGN	0.99876	0.93709	0.96694	22365
DoS	0.91236	0.95708	0.93419	15261
Fuzzy	0.99968	0.97513	0.98725	12745
GEAR	1.00000	0.93729	0.96763	14048
RPM	0.89505	1.00000	0.94462	15581
accuracy			0.95925	80000

macro avg	0.96117	0.96132	0.96013	80000
weighted avg	0.96244	0.95925	0.95970	80000

Results from cell 11

Text output:

Treinando SVM em 100000 amostras de treino.

SVM classification report

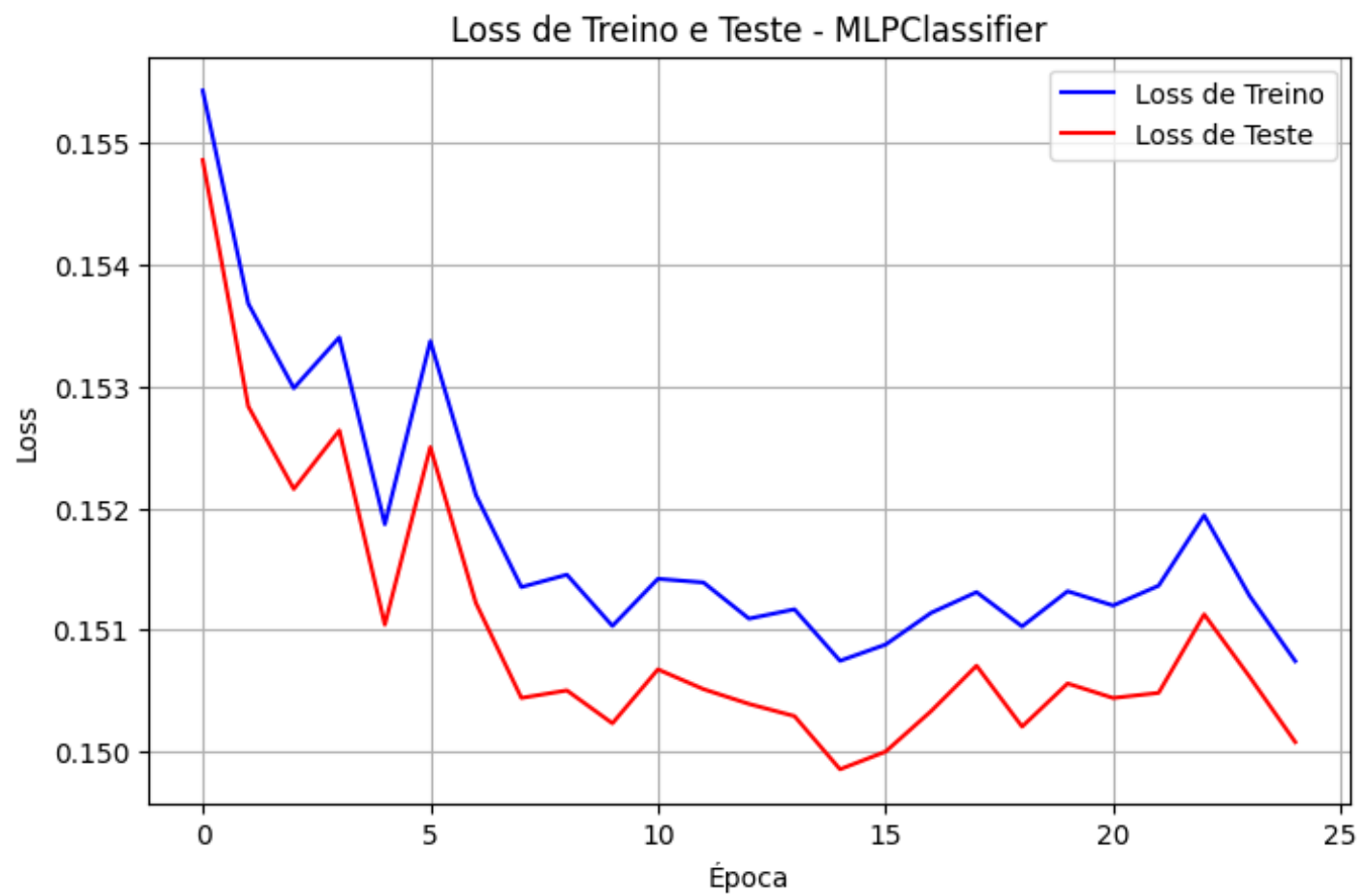
	precision	recall	f1-score	support
BENIGN	0.99139	0.85886	0.92038	200000
DoS	0.73091	0.99617	0.84317	138674
Fuzzy	0.98987	0.92791	0.95789	115937
GEAR	0.99988	0.93642	0.96711	127483
RPM	0.99998	0.94079	0.96948	139275
accuracy			0.92588	721369
macro avg	0.94241	0.93203	0.93161	721369
weighted avg	0.94423	0.92588	0.92930	721369

Results from cell 12

Result:

<Figure size 800x500 with 1 Axes>

Figure:

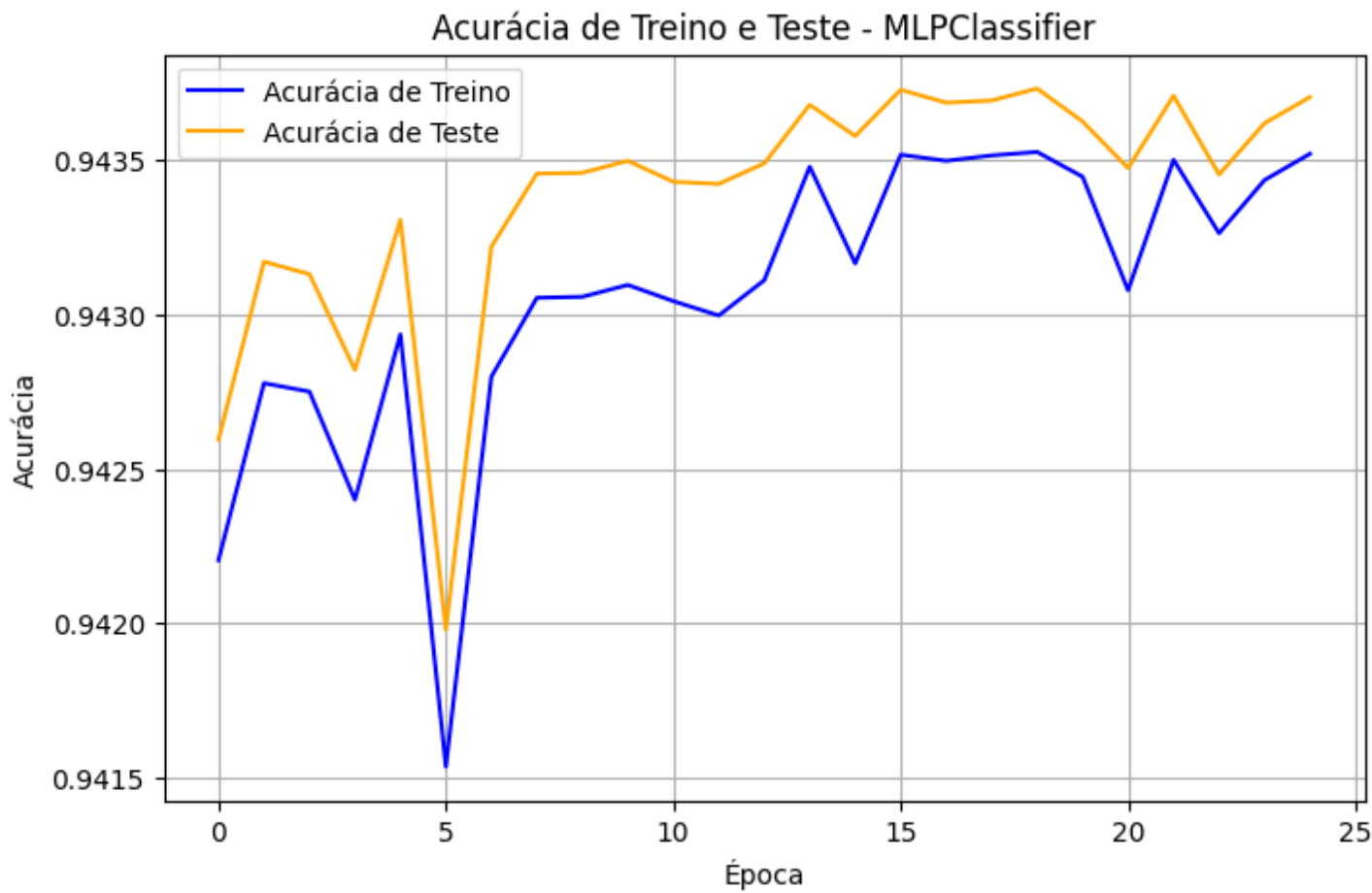


Results from cell 13

Result:

<Figure size 800x500 with 1 Axes>

Figure:



Results from cell 14

Text output:

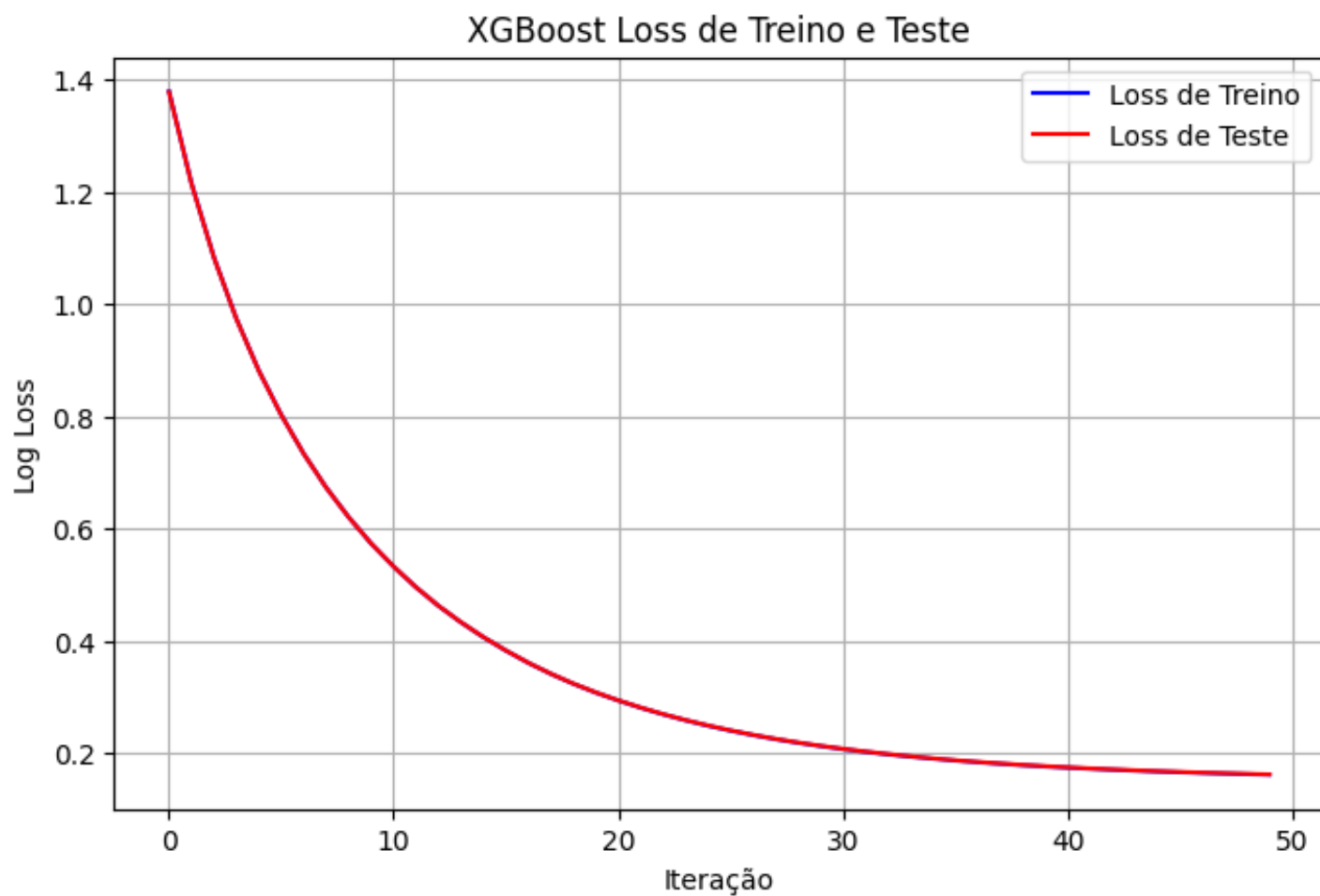
XGBoost j? treinado. Vari?veis dispon?veis: xgb_model, xgb_pred, xgb_proba

Results from cell 15

Result:

<Figure size 800x500 with 1 Axes>

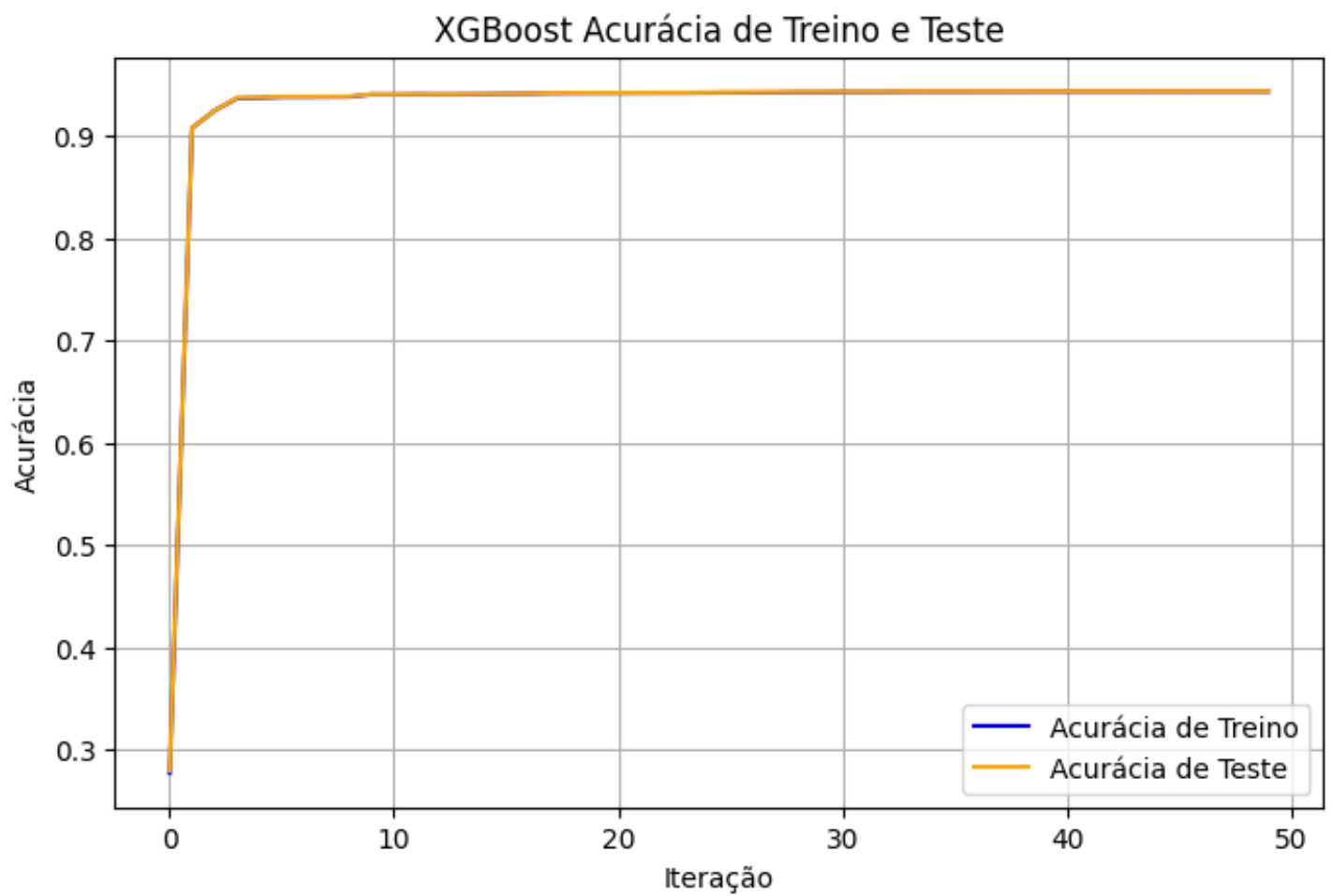
Figure:



Result:

<Figure size 800x500 with 1 Axes>

Figure:



Results from cell 16

Text output:

SVM j? treinado. Vari?veis dispon?veis: svm_model, svm_pred, svm_proba

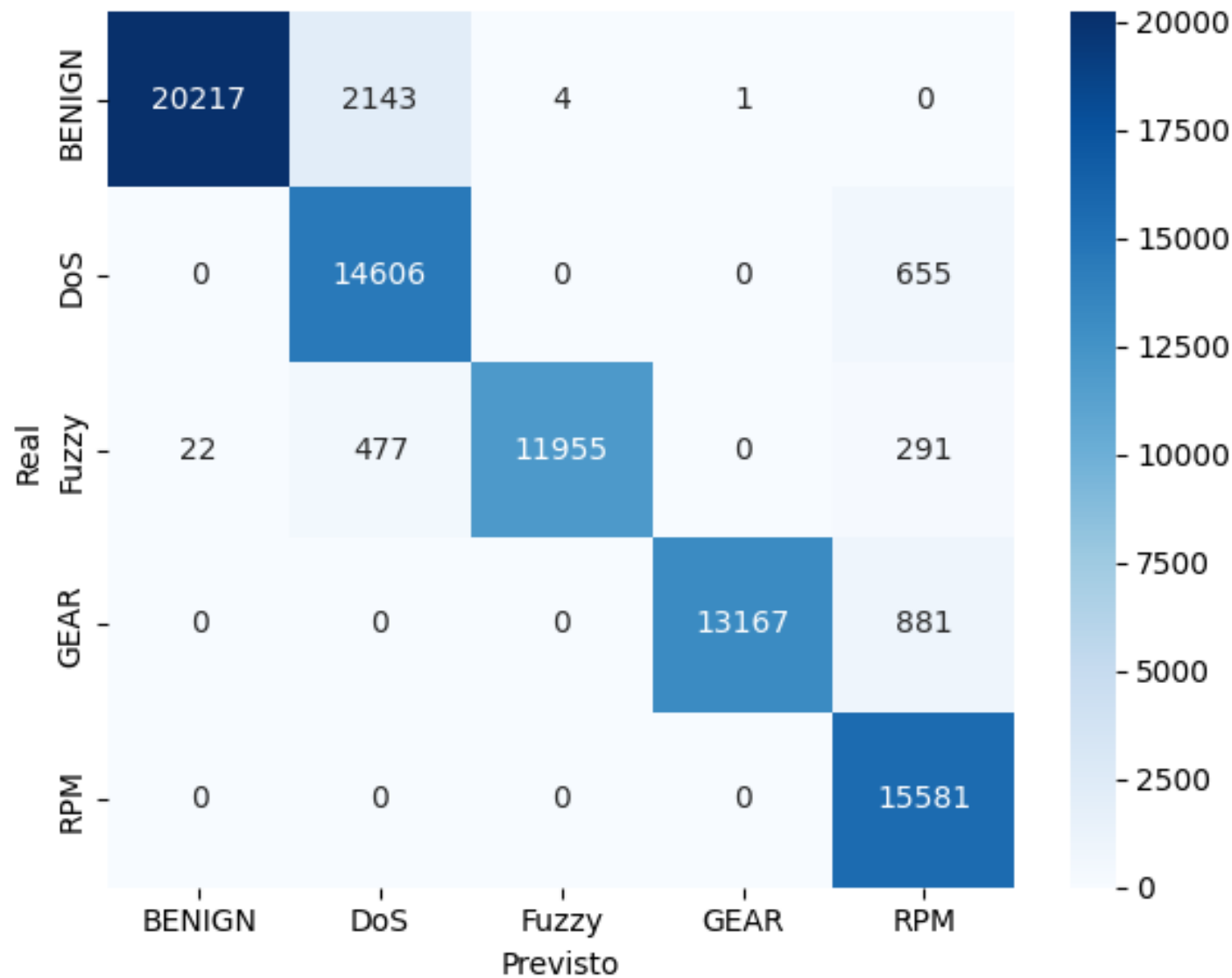
Results from cell 17

Result:

<Figure size 600x500 with 2 Axes>

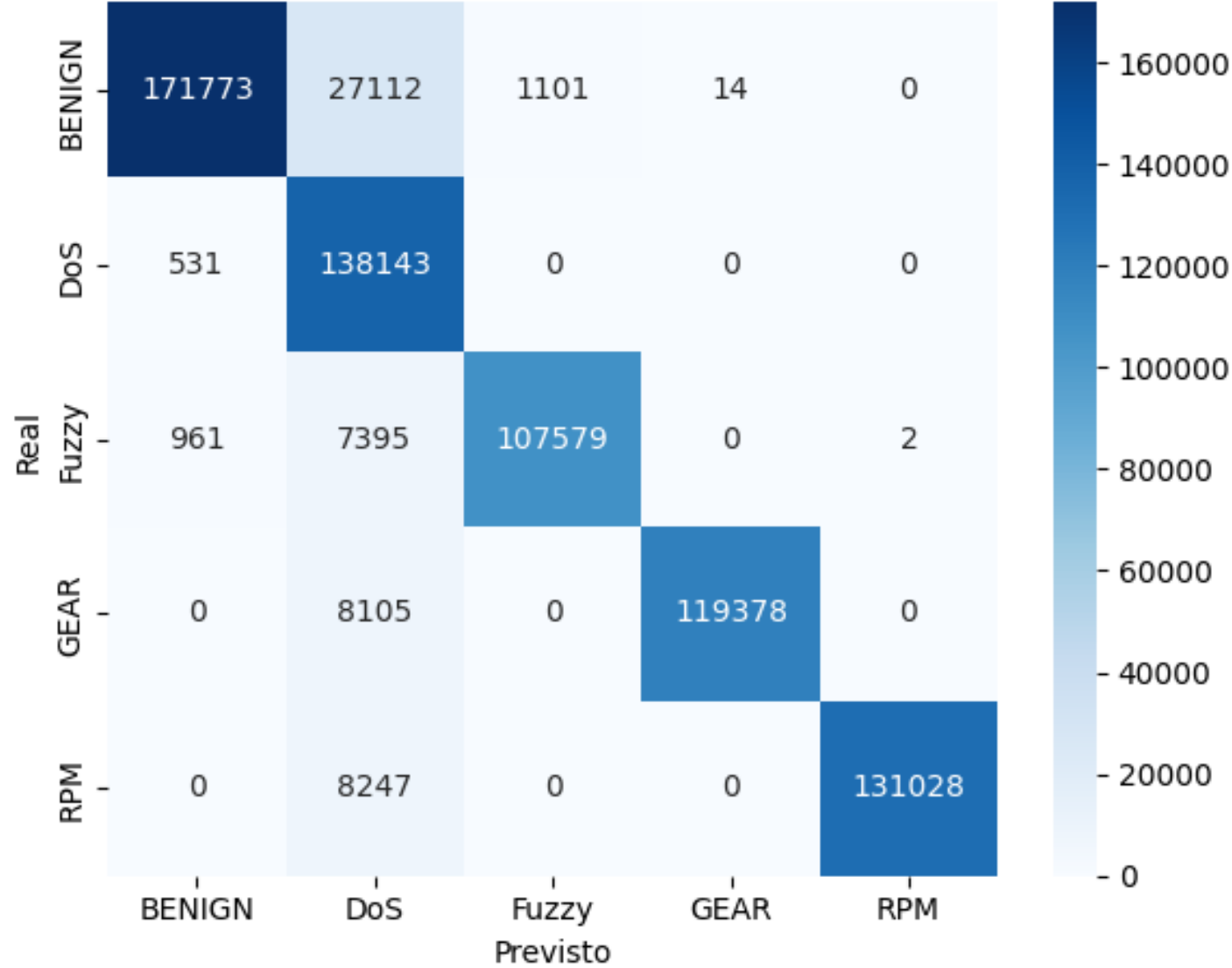
Figure:

Matriz de Confusão - XGBoost



Result:
<Figure size 600x500 with 2 Axes>
Figure:

Matriz de Confusão - SVM



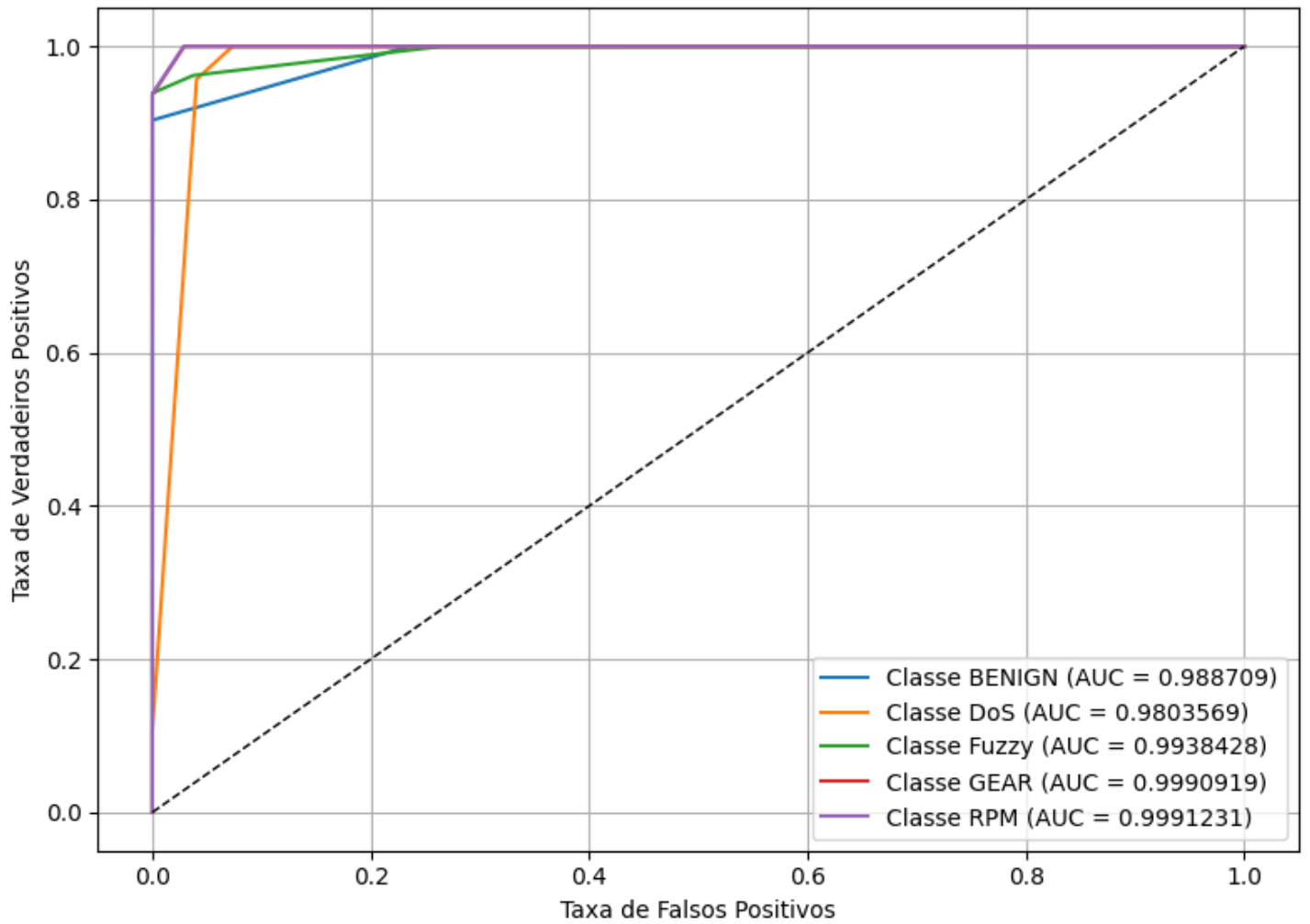
Results from cell 18

Result:

<Figure size 800x600 with 1 Axes>

Figure:

Curva ROC - XGBoost

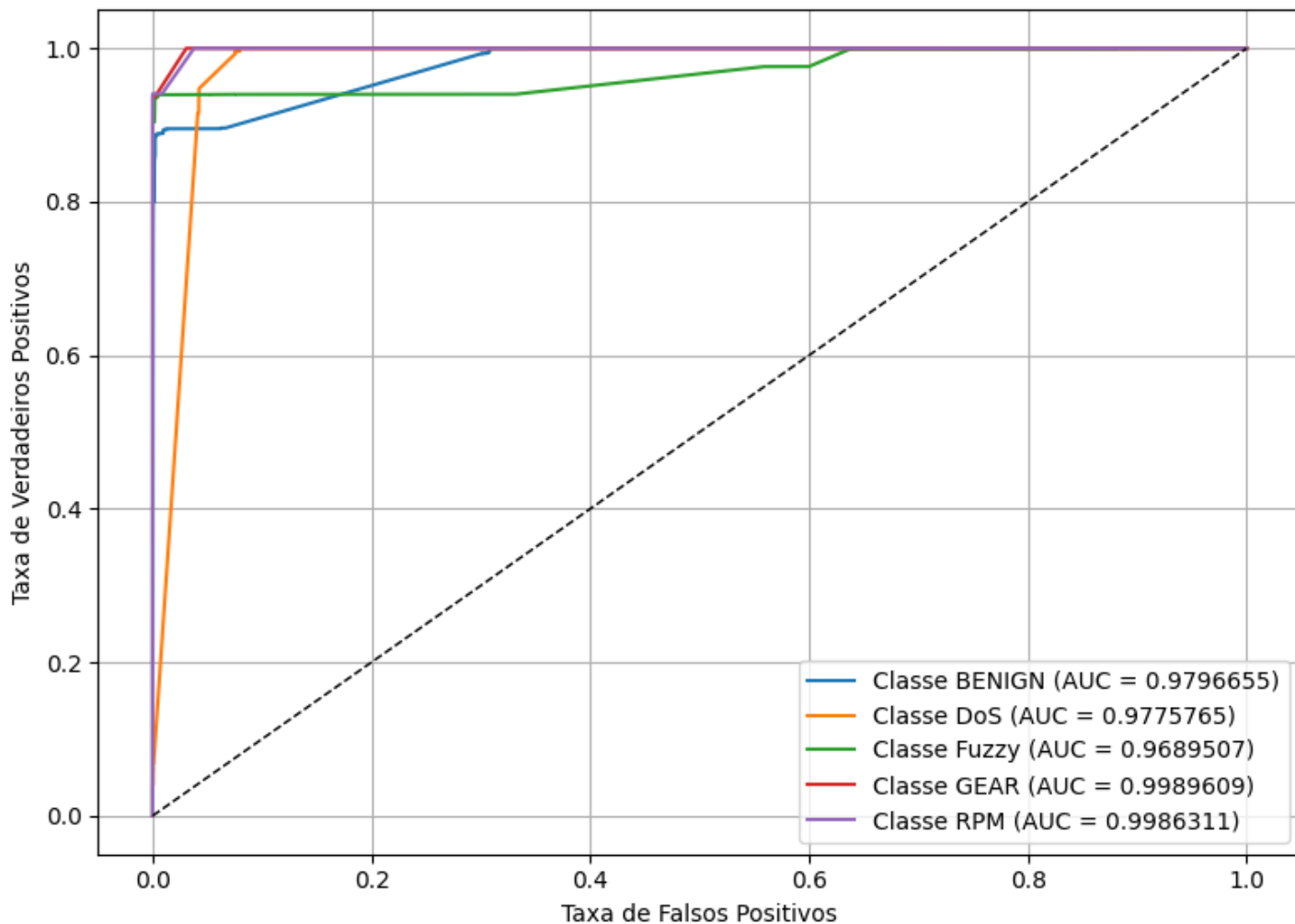


Result:

<Figure size 800x600 with 1 Axes>

Figure:

Curva ROC - SVM



Results from cell 19

Error output:

ModuleNotFoundError: No module named 'seaborn'

```
[31m----- [39m
[31mModuleNotFoundError [39m          Traceback (most recent call last)
[36mCell [39m [36m [39m [32mIn[10] [39m [32m, line 3 [39m
[32m      1 [39m [38;5;28;01mfrom [39;00m sklearn.metrics [38;5;28;01mimport [39;00m confusion_matrix,
ConfusionMatrixDisplay,
classification_report, roc_curve, auc
[32m      2 [39m [38;5;28;01mfrom [39;00m sklearn.preprocessing [38;5;28;01mimport [39;00m label_binarize
[32m----> [39m [32m3 [39m [38;5;28;01mimport [39;00m seaborn [38;5;28;01mas [39;00m sns
[32m      4 [39m [38;5;66;03m# --- Matriz de Confus?o --- [39;00m
[32m      5 [39m conf_matrix = confusion_matrix(y_test, y_pred)
[32m      6 [39m plt.figure(figsize=( [32m6 [39m, [32m5 [39m))
```

[31mModuleNotFoundError [39m: No module named 'seaborn'

Results from cell 20

Text output:

```
===== DoS =====
false_positives: 12222
true_positives: 132432
actual_class_breakdown
true_label
BENIGN    12222
actual_class_breakdown_pct
true_label
BENIGN    100.0
fp_all_zero_rate: 1.0
fp_id_zero_rate: 1.0
tp_all_zero_rate: 0.887
```

tp_id_zero_rate: 0.8869
top_fp_ids
id
0 12222
top_fp_signatures
id val0 val1 val2 val3 val4 val5 val6 val7 true_label
0 0 0 0 0 0 0 0 0 BENIGN 12222
fp_feature_means
id 0.0
val0 0.0
val1 0.0
val2 0.0
val3 0.0
val4 0.0
val5 0.0
val6 0.0
val7 0.0

===== GEAR =====
false_positives: 17146
true_positives: 127483
actual_class_breakdown
true_label
RPM 8247
DoS 6242
Fuzzy 2657
actual_class_breakdown_pct
true_label
RPM 48.1
DoS 36.4
Fuzzy 15.5
fp_all_zero_rate: 0.0
fp_id_zero_rate: 0.0
tp_all_zero_rate: 0.0
tp_id_zero_rate: 0.0

top_fp_ids
id
1520 17145
1657 1
top_fp_signatures
id val0 val1 val2 val3 val4 val5 val6 val7 true_label
1520 1 0 0 0 0 0 0 0 RPM 8247
DoS 6242
Fuzzy 2656
1657 1 48 23 251 103 28 12 122 Fuzzy 1
fp_feature_means
id 1520.01
val0 1.00
val1 0.00
val2 0.00
val3 0.01
val4 0.01
val5 0.00
val6 0.00
val7 0.01